Switch-mode Power Rectifier

MUR550APFG, MURD550PFG, MUR550PFG, MURF550PFG, NRVUD550PFT4G, NRVUD550PFT4G-VF01

ULTRAFAST RECTIFIER
5.0 AMPERES, 520 VOLTS

These state-of-the-art devices are designed for power factor correction in discontinuous and critical conduction mode.

Features

- 520 V Rating Meets 80% Derating Requirements of Major OEMs
- Low Forward Voltage Drop
- Low Leakage
- Ultrafast 95 Nanosecond Recovery Time
- Reduces Forward Conduction Loss
- NRVUD Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC-Q101 Qualified and PPAP Capable
- These Devices are Pb-Free, Halogen Free/BFR Free and are RoHS Compliant

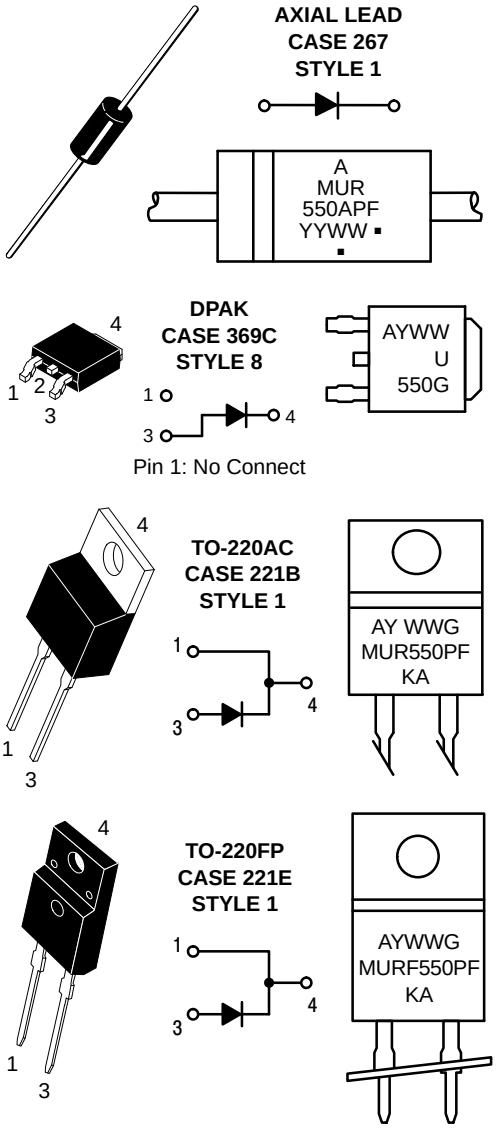
Applications

- DCM PFC Designs
- Switching Power Supplies
- Power Inverters

Mechanical Characteristics:

- Case: Epoxy, Molded
- Epoxy Meets UL 94 V-0 @ 0.125 in
- Weight: MUR550APFG: 1.1 Gram (Approximately)
  MURD550PFG, NRVUD550PFT4G, NRVUD550PFT4G-VF01: 0.4 Gram (Approximately)
  MUR550PFG, MURF550PFG: 1.9 Gram (Approximately)
- Finish: All External Surfaces Corrosion Resistant and Terminal Leads are Readily Solderable
- Lead and Mounting Surface Temperature for Soldering Purposes: 260 °C Max. for 10 Seconds

MARKING DIAGRAMS



A	= Assembly Location*
YY, Y	= Year
WW	= Work Week
▪ or G	= Pb-Free Package
KA	= Diode Polarity

*The Assembly Location Code (A) is front side optional. In cases where the Assembly Location is stamped in the package bottom (molding ejecter pin), the front side assembly code may be blank.

(Note: Microdot may be in either location)

ORDERING INFORMATION

See detailed ordering and shipping information on page 2 of this data sheet.

NOTE: Some of the devices on this data sheet have been **DISCONTINUED**. Please refer to the table on page 2.

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ORDERING INFORMATION

Device	Package	Shipping†
MUR550APFRLG	Axial	1,500 Tape & Reel
MURD550PFT4G	DPAK (Pb-Free)	2,500 Tape & Reel

DISCONTINUED (Note 1)

MUR550APFG	Axial	500 Units/Bag
NRVUD550PFT4G*	DPAK (Pb-Free)	2,500 Tape & Reel
NRVUD550PFT4G-VF01*	DPAK (Pb-Free)	50 Units / Rail
MUR550PFG	TO-220AC (Pb-Free)	50 Units / Rail
MURF550PFG	TO-220FP (Pb-Free)	50 Units / Rail

† For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

* NRVUD Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC-Q101 Qualified and PPAP Capable.

1. **DISCONTINUED:** These devices are not recommended for new design. Please contact your **onsemi** representative for information. The most current information on these devices may be available on www.onsemi.com.

MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Peak Repetitive Reverse Voltage Working Peak Reverse Voltage DC Blocking Voltage	V_{RRM} V_{RWM} V_R	520	V
Average Rectified Forward Current (Rated V_R) T_C = 65 °C MUR550APFG, NRVUD550PFT4G-VF01 (Rated V_R) T_C = 160 °C MURD550PFG, NRVUD550PFT4G, MUR550PFG, MURF550PFG	$I_{F(AV)}$	 5.0 5.0	A
Non-Repetitive Peak Surge Current (Surge Applied at Rated Load Conditions Halfwave, 60 Hz) MUR550APFG NRVUD550PFT4G, NRVUD550PFT4G-VF01, MURD550PFG MUR550PFG, MURF550PFG	I_{FSM}	 85 75 100	A
Operating Junction Temperature Range	T_J	−65 to +175	°C
Storage Temperature Range	T_{stg}	−65 to +175	°C
ESD Ratings: Machine Model = C Human Body Model = 3B	ESD	> 400 > 8000	V

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

THERMAL CHARACTERISTICS

Characteristic	Symbol	Value	Unit
Thermal Resistance, Junction-to-Case (Note 2) MURD550PG, MUR550PFG, NRVUD550PFT4G, NRVUD550PFT4G-VF01 MURF550PFG	$R_{\theta JC}$	 2.8 5.75	°C/W
Thermal Resistance, Junction-to-Ambient MUR550APFG NRVUD550PFT4G, NRVUD550PFT4G-VF01, MURD550PFG (Note 4), MURF550PFG	$R_{\theta JA}$	Note 3 62 75	°C/W

2. Rating applies when surface mounted on the minimum pad sizes recommended.
3. See Note 6, Ambient Mounting Data.
4. 1 inch square pad size on FR4 board.

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ELECTRICAL CHARACTERISTICS

Characteristic	Symbol	Value	Unit
Maximum Instantaneous Forward Voltage Drop (Note 5) (I_F = 5.0 A, T_J = 25 °C) (I_F = 5.0 A, T_J = 150 °C)	V_F	 1.15 0.98	V
Maximum Instantaneous Reverse Current (Note 5) (V_R = 520 V, T_J = 25 °C) (V_R = 520 V, T_J = 150 °C)	I_R	 5.0 400	μA
Maximum Reverse Recovery Time (I_F = 1.0 A, di/dt = 50 A/μs, V_R = 30 V, T_J = 25 °C)	t_{rr}	95	ns

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

5. Pulse Test: Pulse Width = 300 μs, Duty Cycle ≤ 2.0%.

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NOTE 6 — AMBIENT MOUNTING DATA

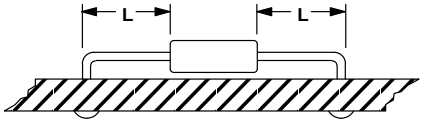
Data shown for thermal resistance junction-to-ambient ($R_{\theta JA}$) for the mountings shown is to be used as typical guideline values for preliminary engineering or in case the tie point temperature cannot be measured.

TYPICAL VALUES FOR $R_{\theta JA}$ IN STILL AIR

Mounting Method		Lead Length, L (IN)				Units
		1/8	1/4	1/2	3/4	
1	$R_{\theta JA}$	50	51	53	55	°C/W
2		58	59	61	63	°C/W
3		28				°C/W

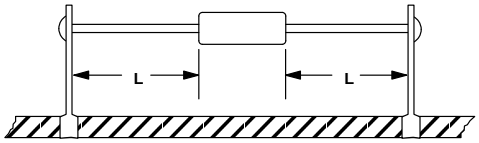
MOUNTING METHOD 1

P.C. Board Where Available Copper Surface area is small.



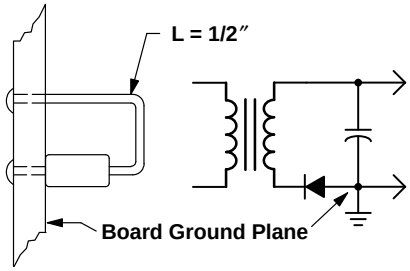
MOUNTING METHOD 2

Vector Push-In Terminals T-28



MOUNTING METHOD 3

P.C. Board with 1−1/2″ x 1−1/2″ Copper Surface



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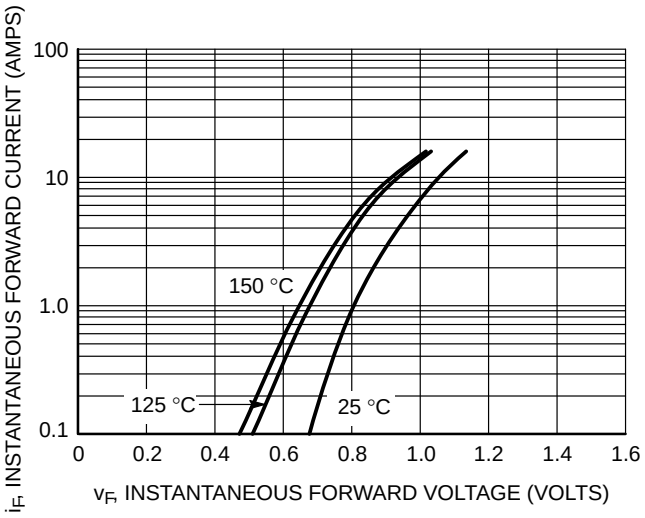


Figure 1. Typical Forward Voltage

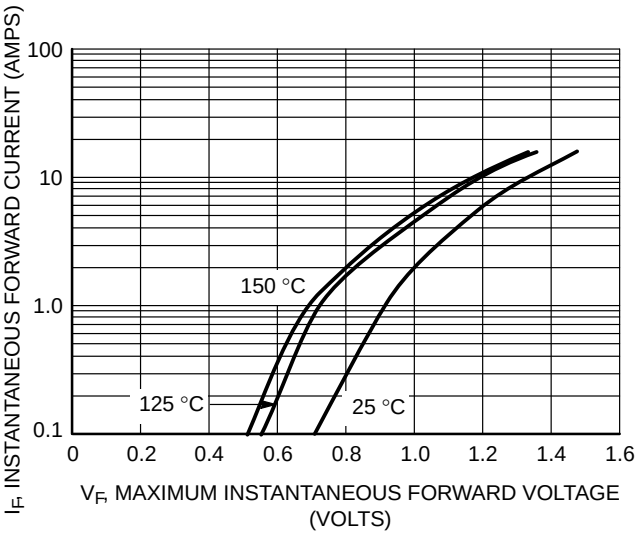


Figure 2. Maximum Forward Voltage

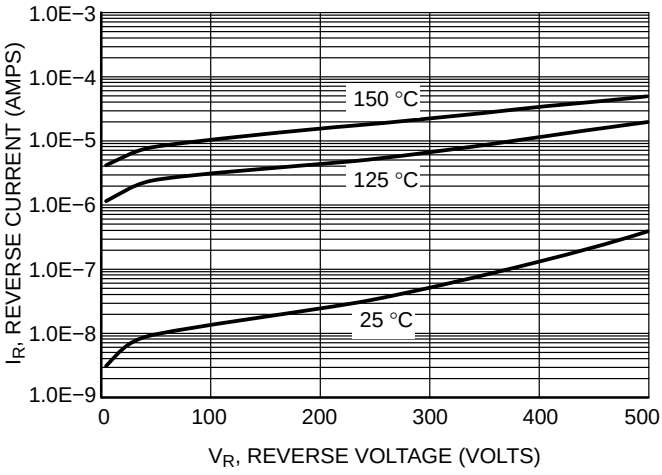


Figure 3. Typical Reverse Current

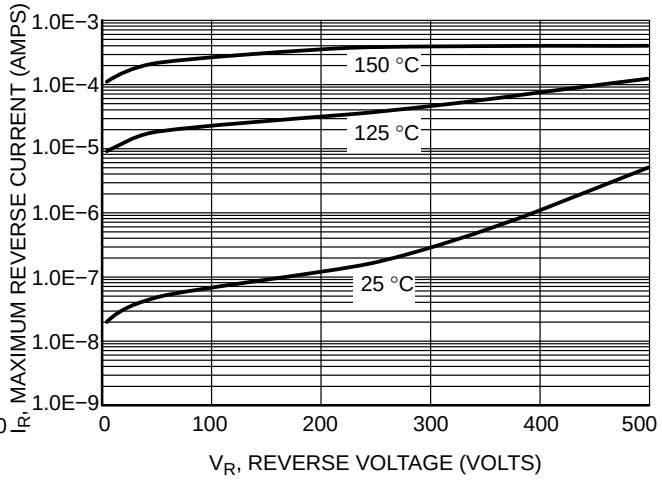


Figure 4. Maximum Reverse Current

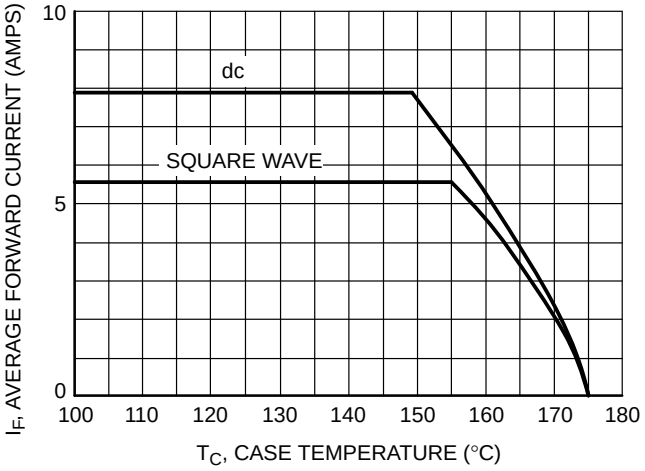


Figure 5. Current Derating

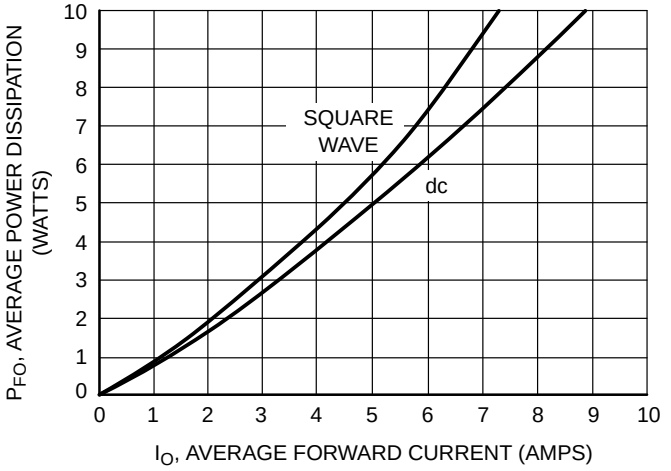


Figure 6. Forward Power Dissipation

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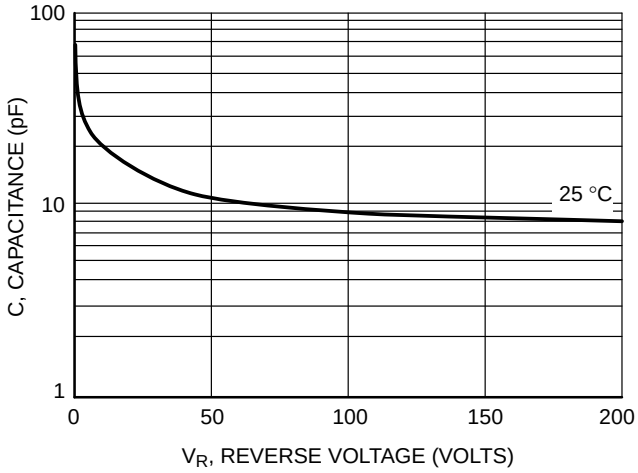


Figure 7. Capacitance

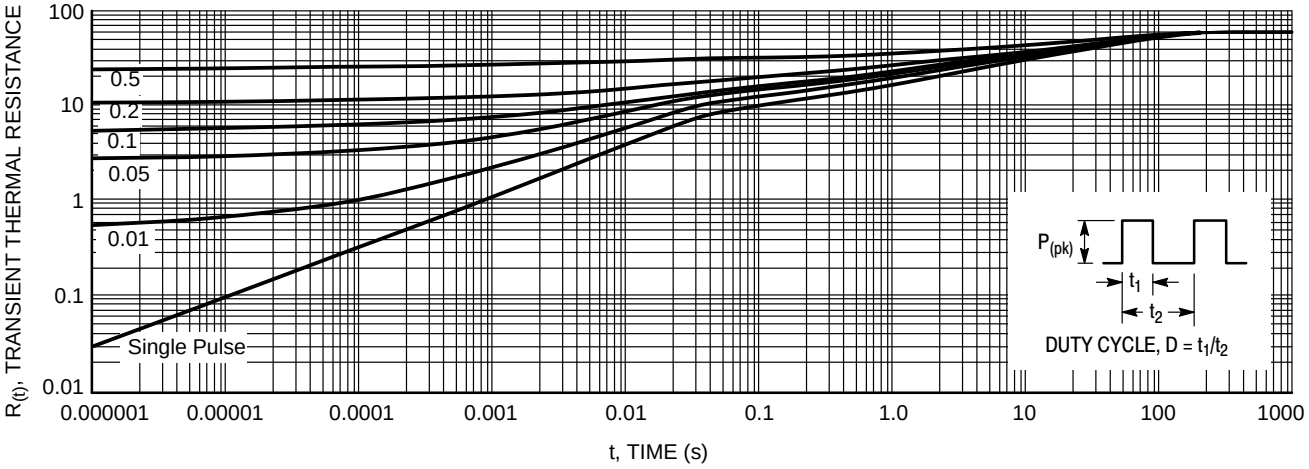


Figure 8. Thermal Response for MUR550APFG

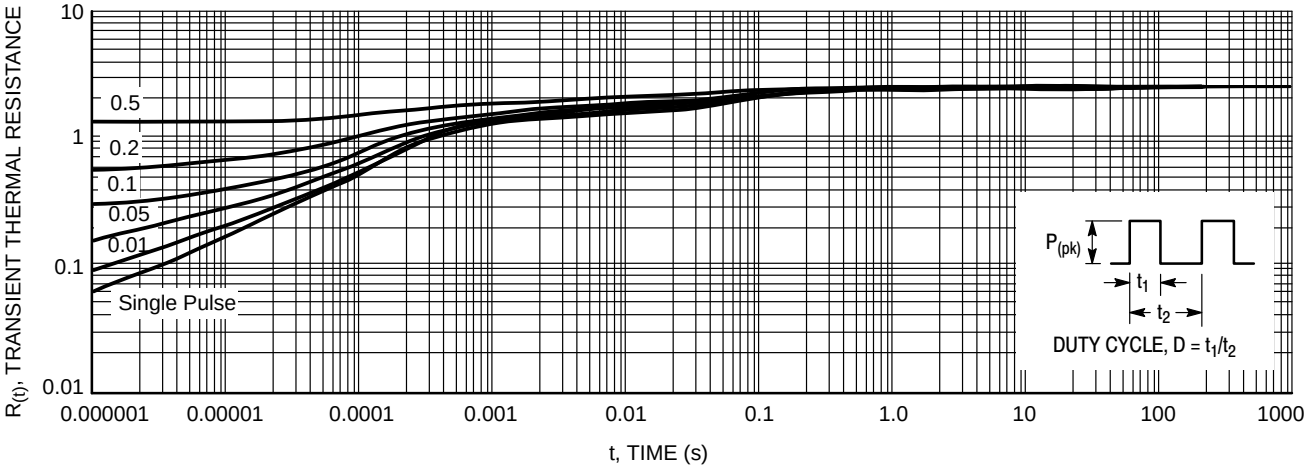


Figure 9. Thermal Response for MURD550PFG, NRVUD550PFT4G, NRVUD550PFT4G-VF01

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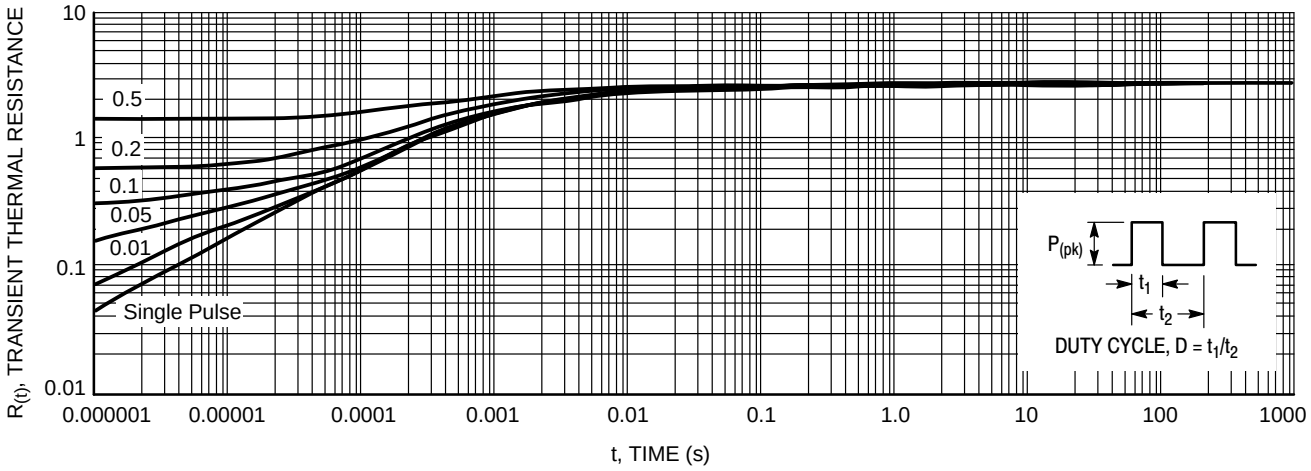


Figure 10. Thermal Response for MUR550PFG

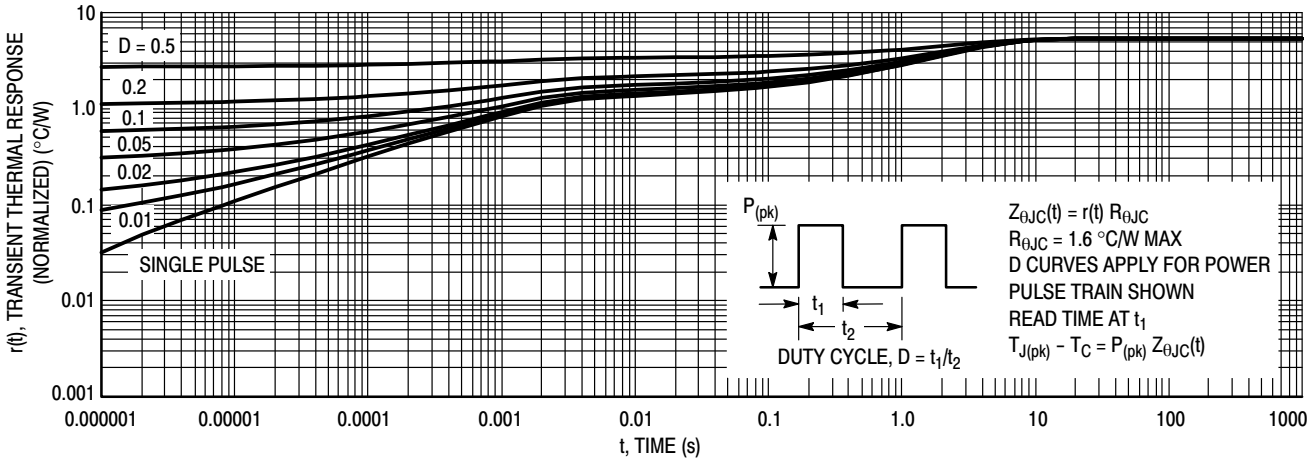


Figure 11. Thermal Response, (MURF550PFG) Junction-to-Case ($R_{\theta JC}$)

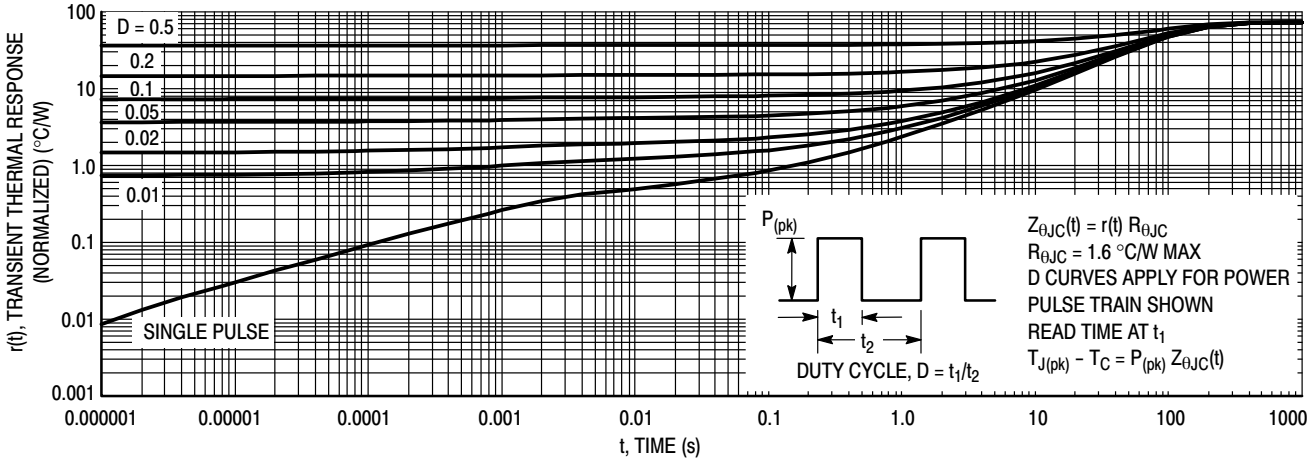


Figure 12. Thermal Response, (MURF550PFG) Junction-to-Ambient ($R_{\theta JA}$)

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REVISION HISTORY

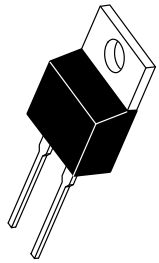
Revision	Description of Changes	Date
12	Rebranded the Data Sheet to **onsemi** format. MUR550APFG, NRVUD550PFT4G, NRVUD550PFT4G−VF01, MUR550PFG, MURF550PFG OPNs marked as Discontinued.	10/13/2025

This document has undergone updates prior to the inclusion of this revision history table. The changes tracked here only reflect updates made on the noted approval dates.

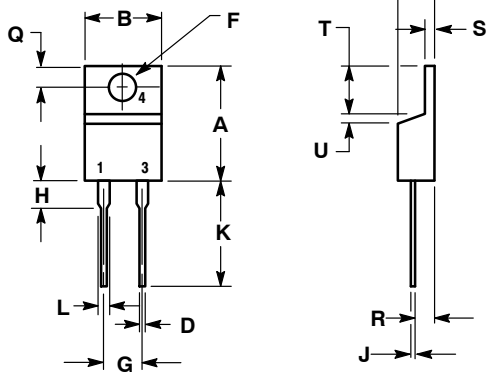
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TO−220, 2−LEAD
CASE 221B−04
ISSUE F

DATE 12 APR 2013



SCALE 1:1



NOTES:
1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: INCH.

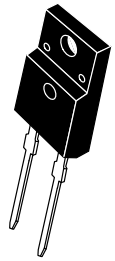
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DIM	MIN	MAX	MIN	MAX
A	0.595	0.620	15.11	15.75
B	0.380	0.405	9.65	10.29
C	0.160	0.190	4.06	4.82
D	0.025	0.039	0.64	1.00
F	0.142	0.161	3.61	4.09
G	0.190	0.210	4.83	5.33
H	0.110	0.130	2.79	3.30
J	0.014	0.025	0.36	0.64
K	0.500	0.562	12.70	14.27
L	0.045	0.060	1.14	1.52
Q	0.100	0.120	2.54	3.04
R	0.080	0.110	2.04	2.79
S	0.045	0.055	1.14	1.39
T	0.235	0.255	5.97	6.48
U	0.000	0.050	0.000	1.27

STYLE 1:
PIN 1. CATHODE
2. N/A
3. ANODE
4. CATHODE

STYLE 2:
PIN 1. ANODE
2. N/A
3. CATHODE
4. ANODE

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DESCRIPTION:	**TO−220, 2−LEAD**		**PAGE 1 OF 1**

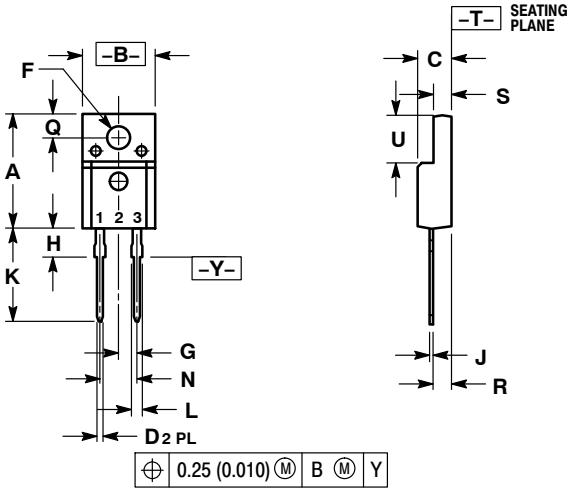
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SCALE 1:1

TO−220 FULLPAK, 2−LEAD
CASE 221E−01
ISSUE A

DATE 21 JAN 2008

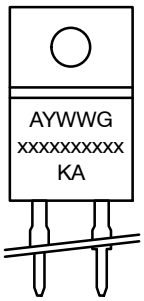


NOTES:
1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: INCH.

	INCHES		MILLIMETERS	
DIM	MIN	MAX	MIN	MAX
A	0.617	0.633	15.67	16.07
B	0.392	0.408	9.96	10.36
C	0.177	0.193	4.50	4.90
D	0.024	0.039	0.60	1.00
F	0.121	0.129	3.08	3.28
G	0.100 BSC		2.54 BSC	
H	0.117	0.133	2.98	3.38
J	0.018	0.025	0.45	0.64
K	0.499	0.562	12.68	14.27
L	0.045	0.060	1.14	1.52
N	0.200 BSC		5.08 BSC	
Q	0.122	0.138	3.10	3.50
R	0.101	0.117	2.56	2.96
S	0.092	0.108	2.34	2.74
U	0.255	0.271	6.48	6.88

STYLE 1:
PIN 1. CATHODE
2. N/A
3. ANODE

GENERIC MARKING DIAGRAM*



Rectifier

A	= Assembly Location
Y	= Year
WW	= Work Week
G	= Pb−Free Package
xxxxxx	= Device Code
KA	= Polarity Designator

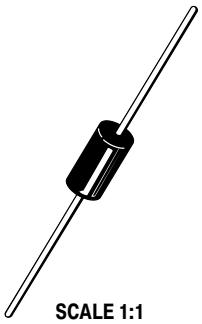
*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “ ▪”, may or may not be present.

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DESCRIPTION:	TO−220 FULLPAK, 2−LEAD	PAGE 1 OF 1

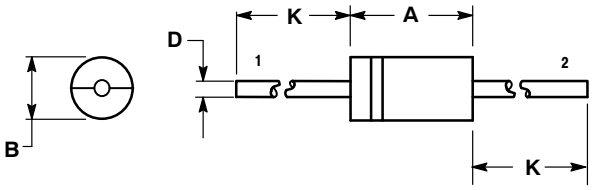
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AXIAL LEAD
CASE 267−05
ISSUE G

DATE 06 JUN 2000



SCALE 1:1



NOTES:
1. DIMENSIONS AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: INCH.
3. 267−04 OBSOLETE, NEW STANDARD 267−05.

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DIM	MIN	MAX	MIN	MAX
A	0.287	0.374	7.30	9.50
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D	0.047	0.051	1.20	1.30
K	1.000	---	25.40	---

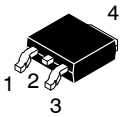
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2. ANODE

STYLE 2:
NO POLARITY

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DESCRIPTION:	**AXIAL LEAD**		**PAGE 1 OF 1**

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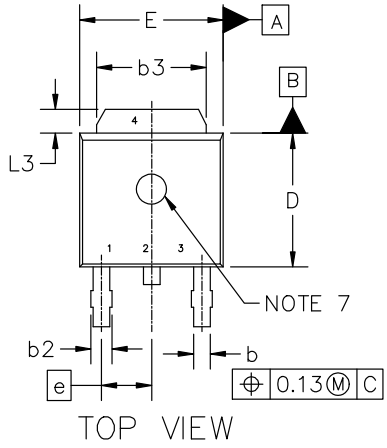
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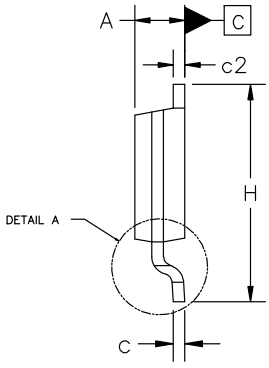
SCALE 1:1

DPAK−3 6.10x6.54x2.28, 2.29P
CASE 369C
ISSUE L

DATE 24 AUG 2026

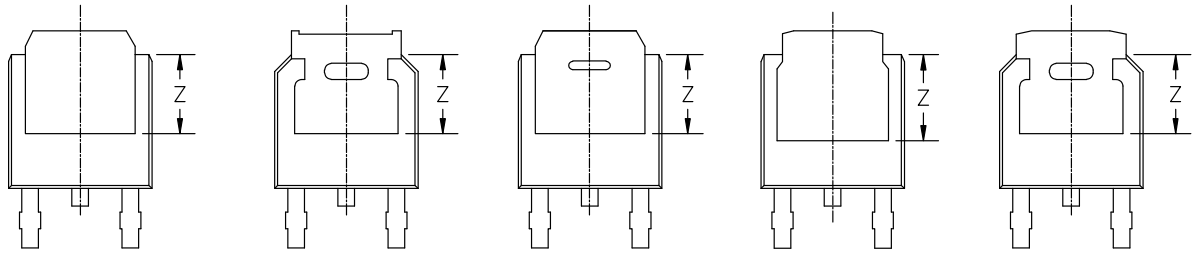


TOP VIEW



SIDE VIEW

MILLIMETERS			
DIM	MIN	NOM	MAX
A	2.18	2.28	2.38
A1	0.00	−−−	0.13
b	0.63	0.76	0.89
b2	0.72	0.93	1.14
b3	4.57	5.02	5.46
c	0.46	0.54	0.61
c2	0.46	0.54	0.61
D	5.97	6.10	6.22
E	6.35	6.54	6.73
e	2.29 BSC		
H	9.40	9.91	10.41
L	1.40	1.59	1.78
L1	2.90 REF		
L2	0.51 BSC		
L3	0.89	−−−	1.27
L4	−−−	−−−	1.01
Z	3.93	−−−	−−−

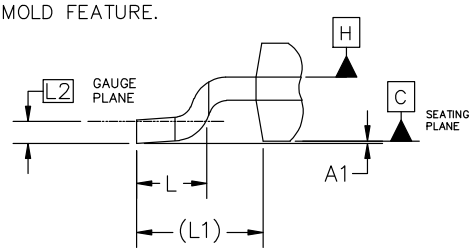


BOTTOM VIEW

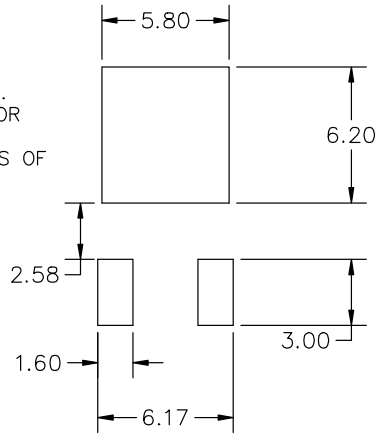
ALTERNATE CONSTRUCTIONS

NOTES:

1. DIMENSIONING AND TOLERANCING ASME Y14.5M, 2018.
2. CONTROLLING DIMENSION: MILLIMETERS.
3. THERMAL PAD CONTOUR OPTIONAL WITHIN DIMENSIONS b3, L3, AND Z.
4. DIMENSIONS D AND E DO NOT INCLUDE MOLD FLASH, PROTRUSIONS, OR BURRS.
5. DIMENSIONS D AND E ARE DETERMINED AT THE OUTERMOST EXTREMES OF THE PLASTIC BODY.
6. DATUMS A AND B ARE DETERMINED AT DATUM PLANE H.
7. OPTIONAL MOLD FEATURE.



DETAIL A
ROTATED 90° CW



RECOMMENDED MOUNTING FOOTPRINT*
*For additional information on our Pb−Free strategy and soldering details, please download the onsemi Soldering and Mounting Techniques reference manual, SOLDERRM/D.

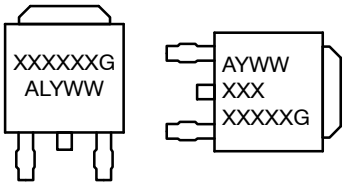
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DESCRIPTION:	**DPAK−3 6.10x6.54x2.28, 2.29P**	**PAGE 1 OF 2**

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DPAK−3 6.10x6.54x2.28, 2.29P
CASE 369C
ISSUE L

DATE 24 AUG 2026

GENERIC MARKING DIAGRAM*



IC **Discrete**

XXXXXX	= Device Code
A	= Assembly Location
L	= Wafer Lot
Y	= Year
WW	= Work Week
G	= Pb−Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “▪”, may or may not be present. Some products may not follow the Generic Marking.

STYLE 1:
PIN 1. BASE
2. COLLECTOR
3. EMITTER
4. COLLECTOR

STYLE 2:
PIN 1. GATE
2. DRAIN
3. SOURCE
4. DRAIN

STYLE 3:
PIN 1. ANODE
2. CATHODE
3. ANODE
4. CATHODE

STYLE 4:
PIN 1. CATHODE
2. ANODE
3. GATE
4. ANODE

STYLE 5:
PIN 1. GATE
2. ANODE
3. CATHODE
4. ANODE

STYLE 6:
PIN 1. MT1
2. MT2
3. GATE
4. MT2

STYLE 7:
PIN 1. GATE
2. COLLECTOR
3. EMITTER
4. COLLECTOR

STYLE 8:
PIN 1. N/C
2. CATHODE
3. ANODE
4. CATHODE

STYLE 9:
PIN 1. ANODE
2. CATHODE
3. RESISTOR ADJUST
4. CATHODE

STYLE 10:
PIN 1. CATHODE
2. ANODE
3. CATHODE
4. ANODE

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